DEPARTMENT OF
SCIENCE & TECHNOLOGY
Ministry of Science and Technology
Government of India

National Quantum Mission

NATIONAL QUANTUM MISSION

Dr. JBV Reddy
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Govt. of India

NATIONAL QUANTUM MISSION (NQM)

- Quantum Technology is one of the 9 missions of national importance, being driven by the **Prime Minister's Science Technology Innovation Advisory Council (PM-STIAC)** for leveraging cutting edge scientific research.

- **Department of Science and Technology** is **implementing the Mission along with other key Ministries.**

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NATIONAL QUANTUM MISSION (NQM)

National Quantum Mission

Quantum Technologies

Communication Sector
Data Security, terrestrial information transmission

Finance Sector
Enhance the security, increase the speed of big data analysis

Health Sector
Less invasive, more accurate Diagnostics, Increase speed of drug discovery

Agriculture
Genetics, production management

Energy Sector
Energy capture, storage, distribution

Space Applications
Satellite communication, Navigation

Transportation
Traffic management, logistics optimization

Defence
Advanced surveillance, precise navigation

NQM: KEY TECHNOLOGY DOMAINS

Quantum Computing

- Superconducting Qubits
- Trapped Ion Qubits
- Photonic Qubits
- Neutral atom Qubits
- Semiconductor Spin Qubits
- Topological Qubits
- Nitrogen Vacancy (NV) Centers
- Quantum Annealing, Algorithms & Applications

Quantum Communication

- Quantum Cryptography
- Quantum Repeaters
- Satellite-Based Quantum Communication
- Free-Space Quantum Communication
- Photon Sources and Detectors
- Post Quantum Cryptography

Quantum Sensing & Metrology

- Atomic Magnetometers
- Atomic Clocks and Radars
- Quantum Gravimeters & Inertial Sensors
- Quantum Enhanced Imaging
- Quantum-enhanced Metrology
- NV Centre based sensors

Quantum Materials & Devices

- Topological Insulators
- Superconductors
- Other 2D Materials/spintronic materials etc.
- Quantum Dots

NQM: DELIVERABLES

Quantum Computing

- Develop intermediate-scale quantum computers with 50-1000 physical qubits

Quantum Communication

- Develop secure quantum communication across 2000 km within India (both satellite- and fibre-based),
- Develop multi-node quantum networks

Quantum Sensing & Metrology

- Develop magnetometers with high sensitivity (up to 1 Femto Tesla/sqrt(Hz)
- Develop gravimeters of high sensitivity (>100 nm/s2)
- Develop atomic clocks with 10-19 fractional instability

Quantum Materials & Devices

- Design and synthesis of quantum materials
- Production of devices for enabling quantum technologies

KEY MILESTONES OF NQM

Phase 1
Up to ~3 Yrs.
Prototype Development

- Establish 4 Thematic Hubs (T-Hubs)
- Fund Support to the T-Hubs and Mission Kick-off
- Prototype Development

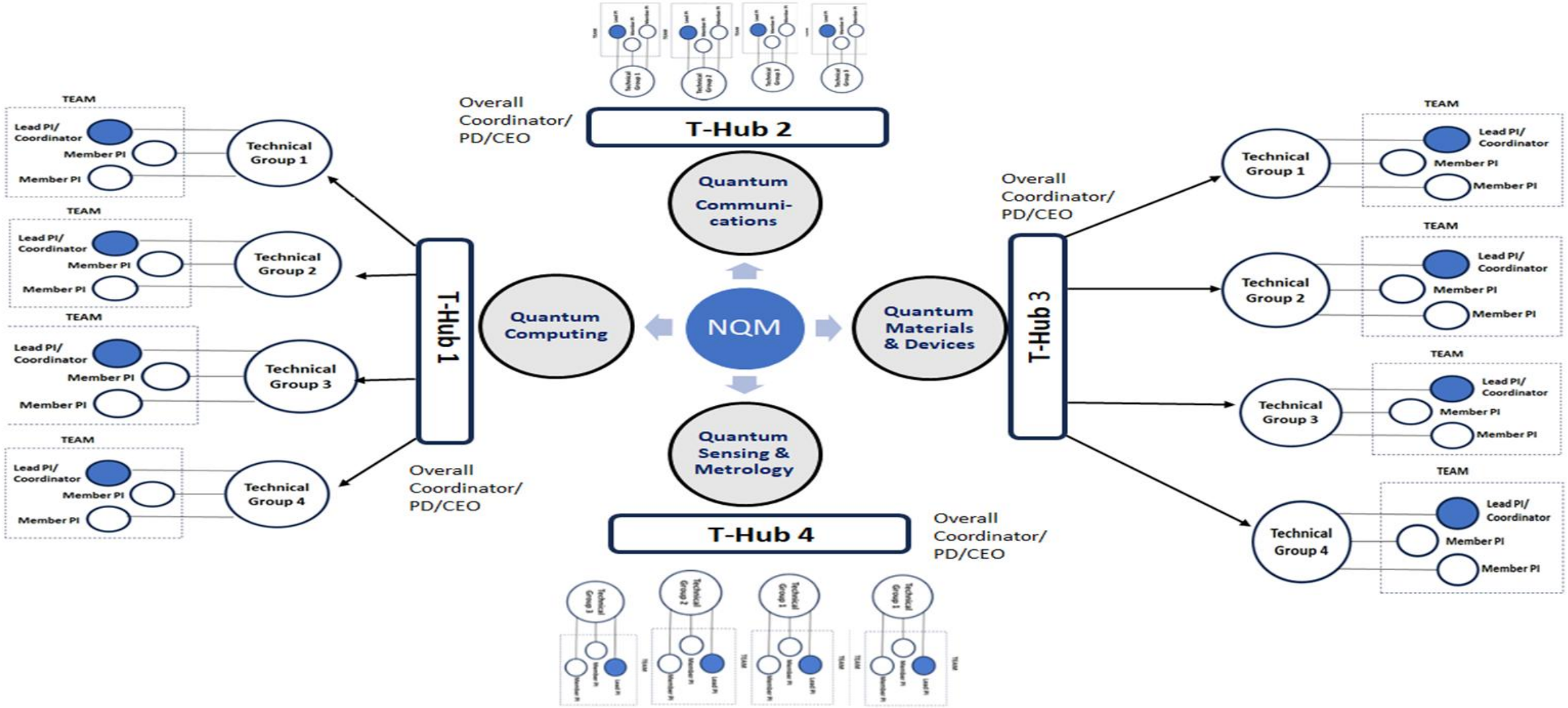
Phase 2
Up to ~ 6 Yrs.
Product Development & Deployment

- Product Development
- Product Deployment & Sectorial Applications

Phase 3
Up to ~8 yrs.
Business Development & Economic Growth

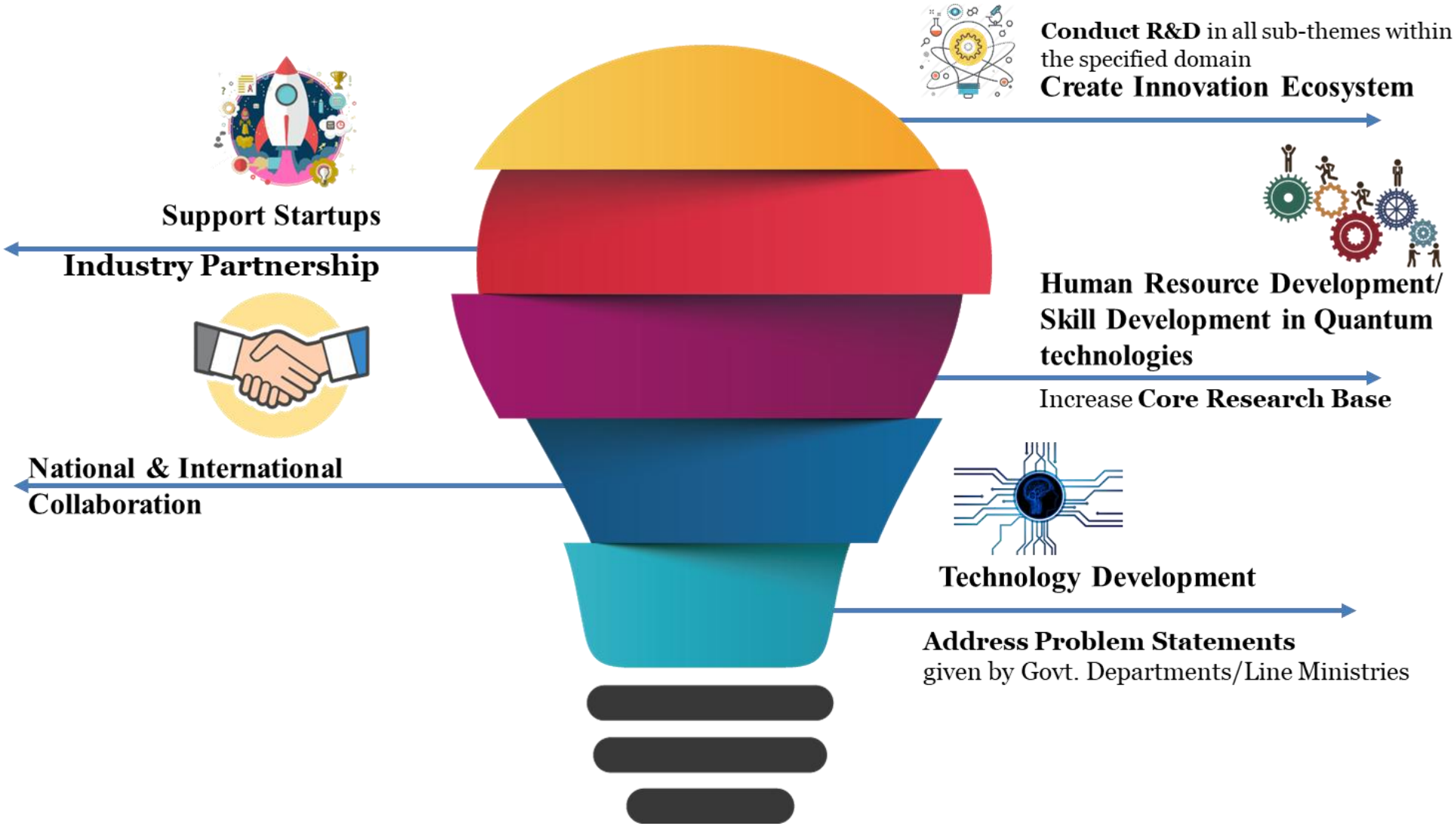
- Sustained Business & Increase of Job Market
- Scale-up Businesses & Contribute to Economic Growth

THEMATIC HUBS (T-HUBS)



T-Hubs are the Section-8 Companies

MANDATE OF T-HUBS



National Quantum Mission

सत्यमेव जयते

विज्ञान और प्रौद्योगिकी विभाग

Department of

SCIENCE & TECHNOLOGY

Thematic Hubs
&
Technical Groups

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QUANTUM COMPUTING

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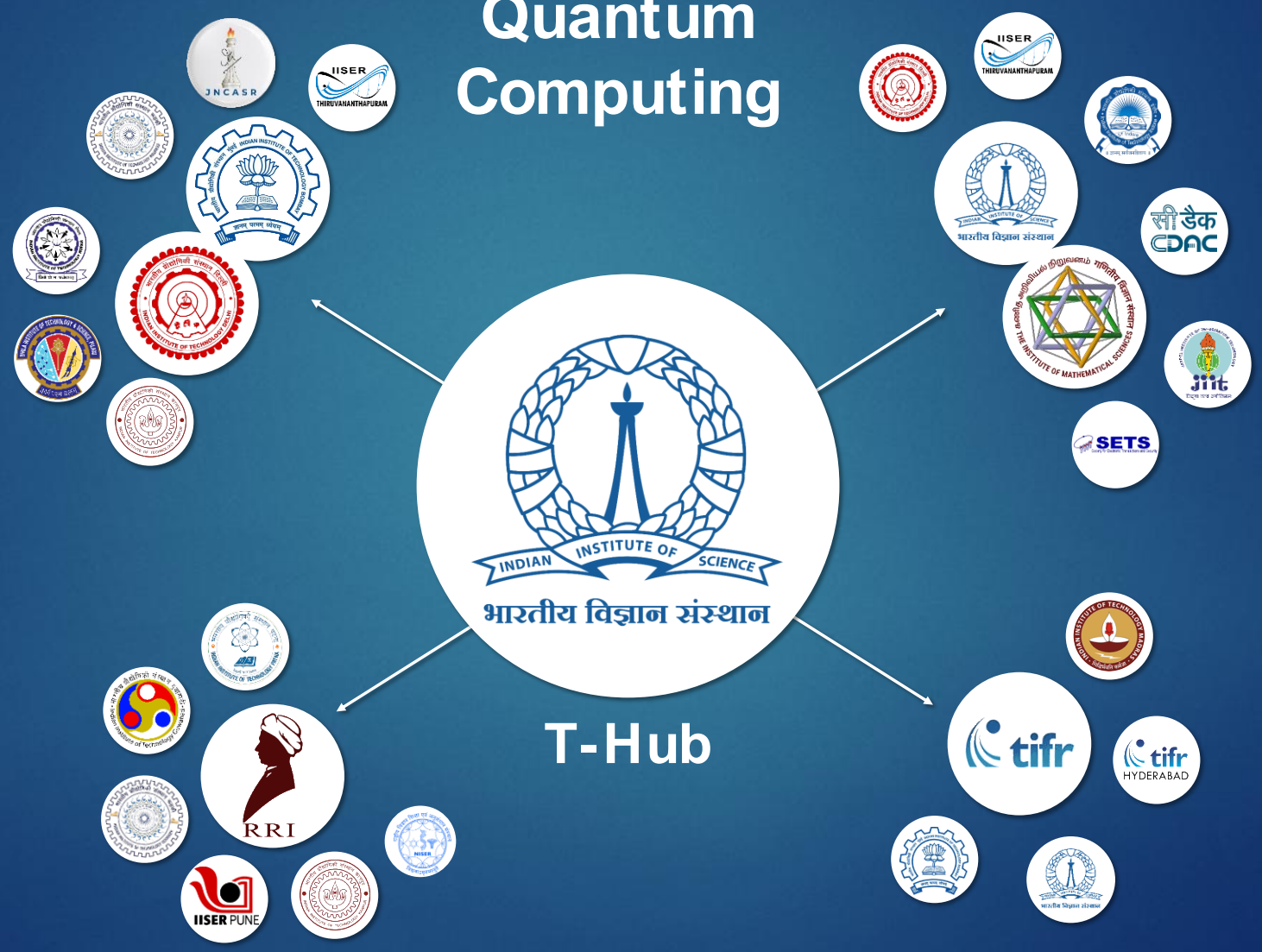

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QUANTUM COMMUNICATIONS

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QUANTUM SENSING & METROLOGY

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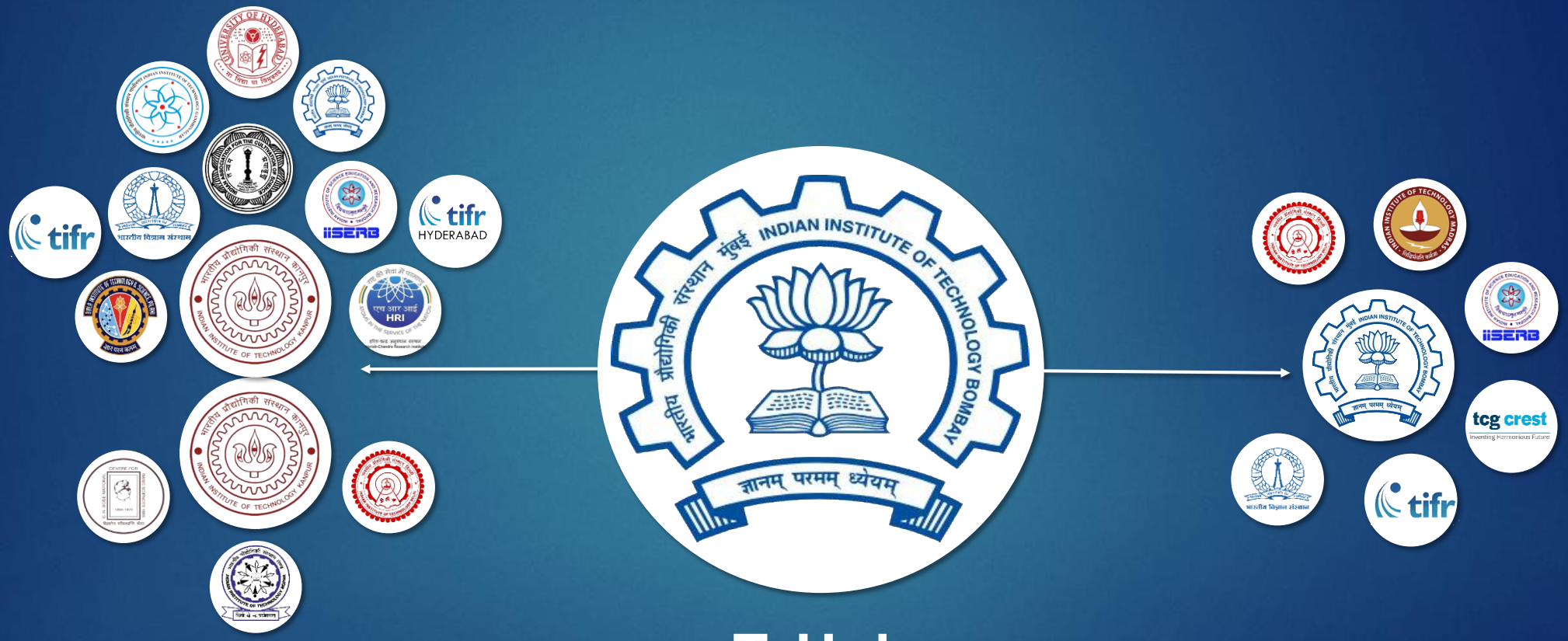

T-Hub

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QUANTUM MATERIALS & DEVICES

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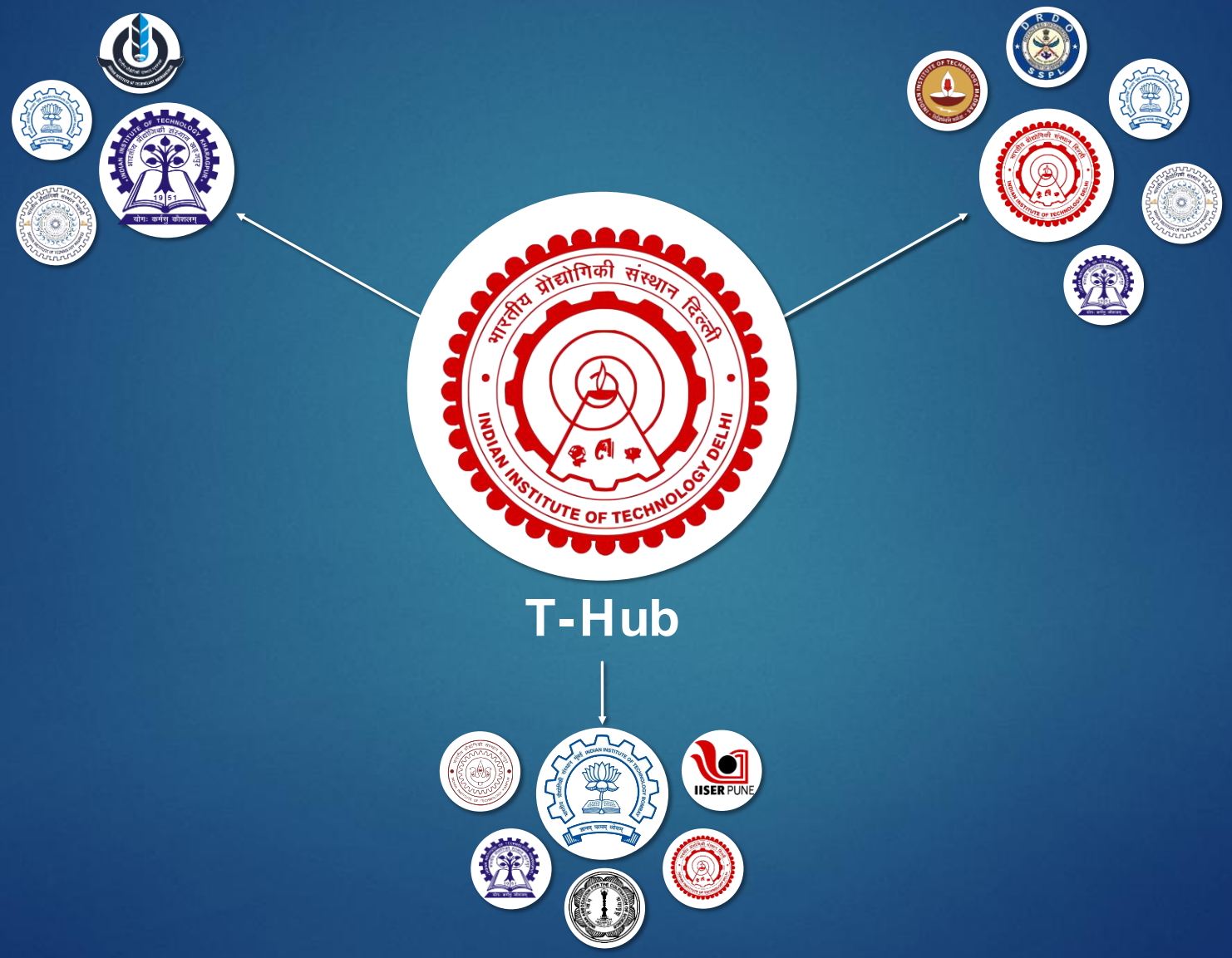

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LAUNCH OF FOUR T-HUBS

The T-Hubs were formally launched on 30th September 2024

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MAPPING OF INSTITUTIONS UNDER 4 T-HUBS

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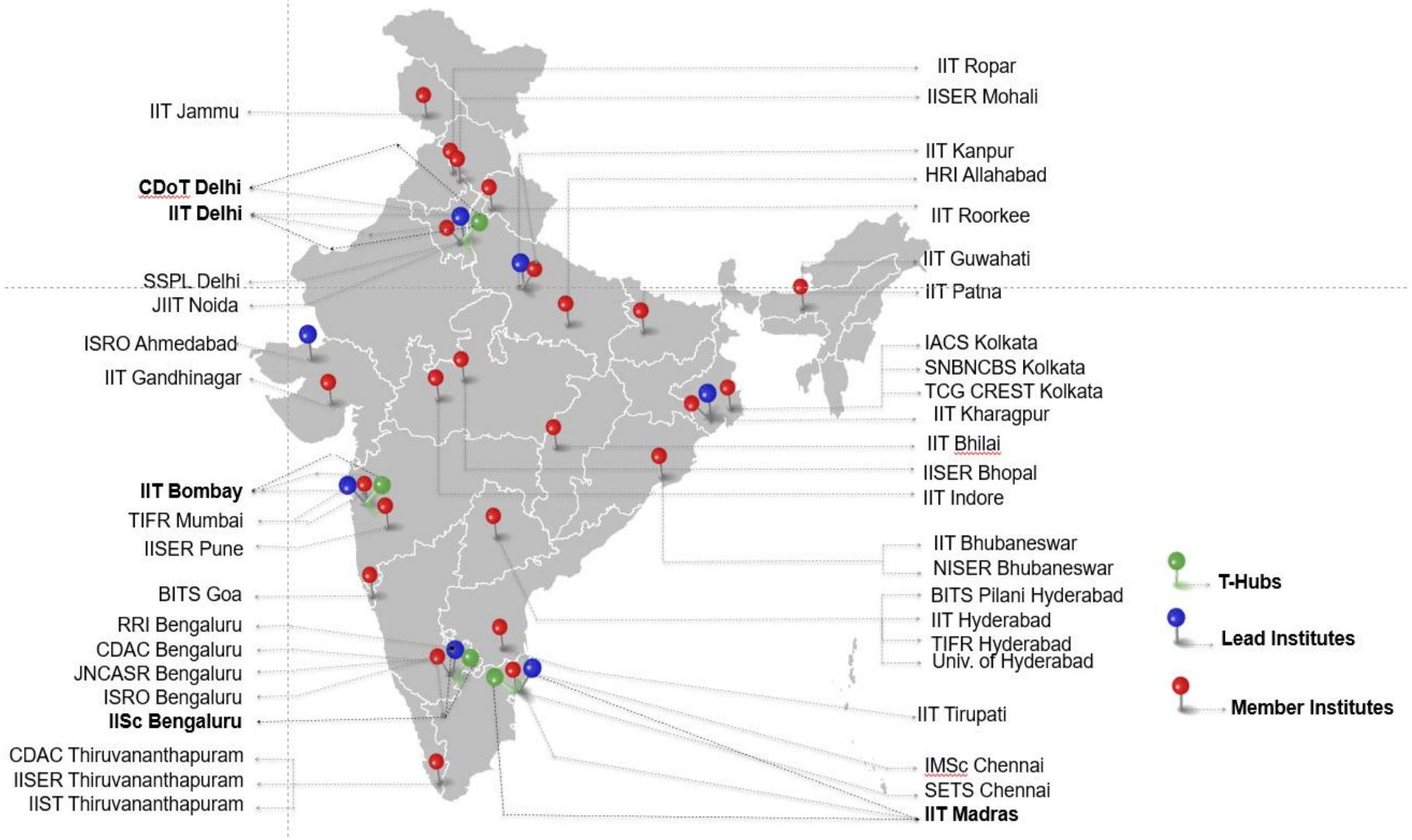
Comprising of 14 Technical Groups (17 Project Teams)



152 researchers from 43 Institutions (Lead & Member) involved in the 4 T-Hubs

Establishment of Fabrication and Central Facilities

State-of-the-art fabrication and centralized facilities are being established at premier institutions.

- The Quantum Computing fabrication facilities at IISc Bengaluru & IIT Bombay will enable the fabrication of superconducting, photonic, and spin qubit chips for scalable quantum architectures.
- The Quantum Sensing & Metrology facilities at IIT Bombay & IIT Kanpur will accelerate breakthroughs in quantum sensing such as advanced quantum sensor platforms for tangible societal and strategic impact.
- The Quantum Materials & Devices fabrication facility at IIT Delhi will drive indigenous quantum materials and device fabrication for scaling up various quantum technologies.

These facilities will enable researchers and startups to develop indigenous prototype quantum processors, sensors, materials, and devices, while also training the nation's human resources in quantum technology hardware, device fabrication, and system integration.

STARTUPS ENGAGEMENT IN NQM

- Allows growth of start-up ecosystem
- Start-ups as active part of technology delivery
- Facilitate Technology transition

T-Hubs

Startup support

T-Hubs create National Facilities hosting sophisticated instruments, manufacturing bases

National facilities created through T-Hub are open to start-ups

T-Hubs will provide technology & business mentoring and equity support to start-ups

STARTUPS ENGAGEMENT IN NQM

- **Guidelines Formulated**
  - Quantum technologies is in nascent stage – requires huge investment
  - Developed exclusive guidelines for effectively on-boarding and supporting startups

Q→NU

QπAI

DIMIRA

PRENISHQ

PRISTINE DIAMONDS PVT. LTD.

QUANASTRA QUANTUM INNOVATIONS

QUPRAYOG

Quan2D Technologies Pvt. Ltd.

8 Startups selected for support

ANNOUNCEMENT OF QUANTUM STARTUPS

Union Minister of State for Science and Technology announced the Quantum Startups

- Launched on 26th November 2024
- Attended by ~400 participants, out of which 200 venture capitalists took part – keen interest in supporting start-ups

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Rolling Call for supporting startups in Quantum Technologies

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Continuous support to early-stage ventures across all the four quantum verticals

Quantum Sensing Technologies

Quantum Computing Technologies

Quantum Materials & Devices

Quantum Secure Communication

INDUSTRY ENGAGEMENT IN NQM

- Technology transition facilitated
- Novel R&D ideas delivered as products

T-Hubs

Industry Engagement

T-Hubs allow industry access to National Facilities

T-Hubs and Industry co-develop technologies with joint IPs

T-Hubs and Industry will develop a revenue-sharing model for deployed technologies

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QUANTUM AWARENESS PROGRAMME

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Enhance awareness of quantum technologies among students, researchers, industry professionals, and policymakers.

Awareness

Build capacity through workshops, training programs, and educational initiatives

Capacity

Contribute to policy development by providing insights for quantum technology regulations and standards

Policy

CAPACITY BUILDING

- NQM in collaboration with AICTE developed a **curriculum on Quantum Technologies for B.Tech. Minor & M.Tech.** These Curriculums were **launched on 24th December 2024 & 28th August 2025** respectively.
- Efforts Underway to **initiate the course**:
  - Course books for the Quantum Curriculum
  - Video lecture series for student learning
  - Faculty Development Programs (FDPs)
- Two dedicated calls have also been launched — one on Quantum Algorithms, to accelerate application-oriented development, and another on Undergraduate Teaching Laboratories in Quantum Technologies, to foster hands-on learning among young talent.

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Thank you

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